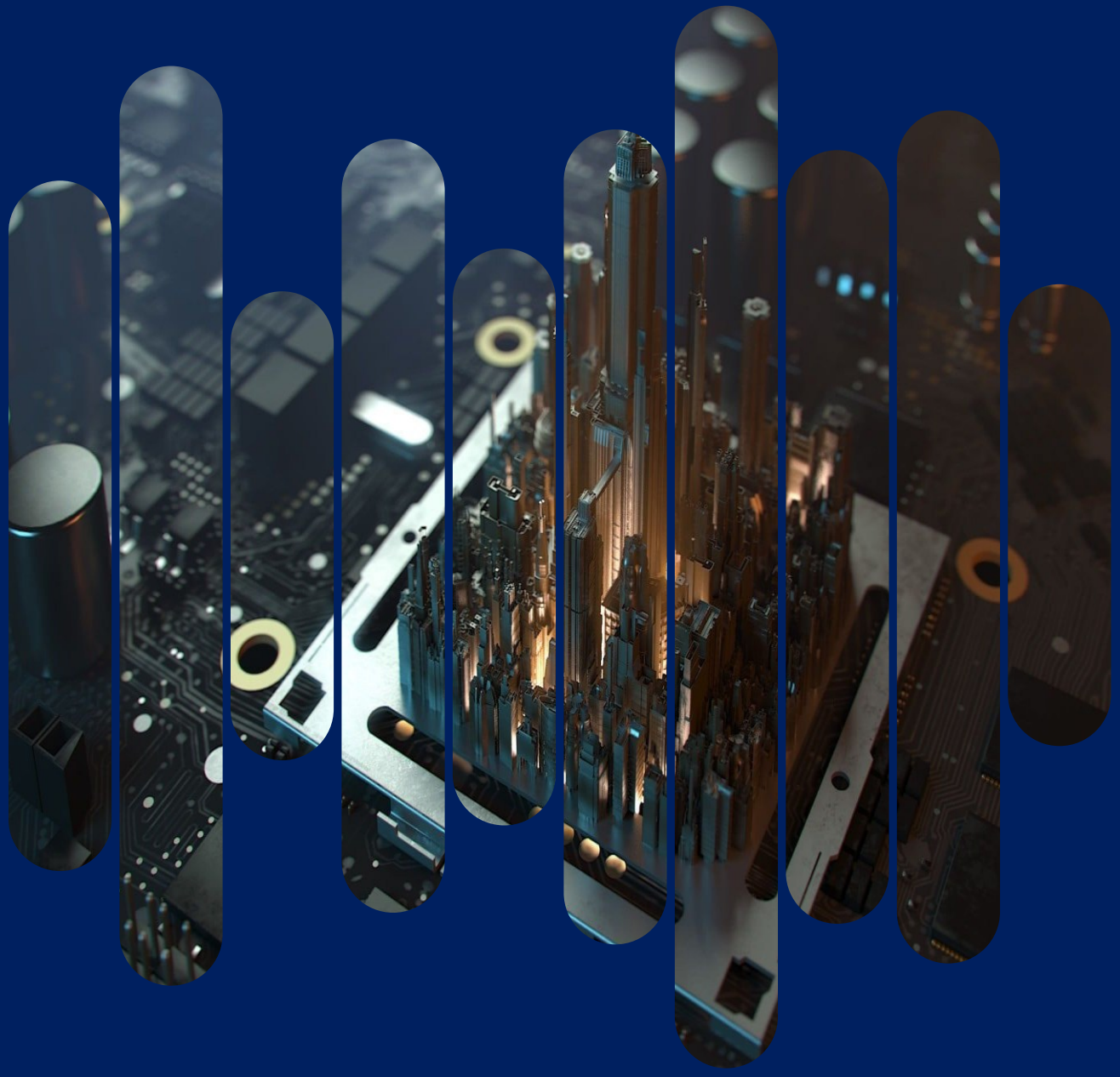
Microprocessor and Computer Architecture



CHAPTER 2

Architecture and Assembly Language Programming of 8085

2.1 Pin Configuration

2.2 Functional Block Diagram

2.3 Intel 8085 Instructions types

2.4 Operation code and operands

2.5 Addressing modes

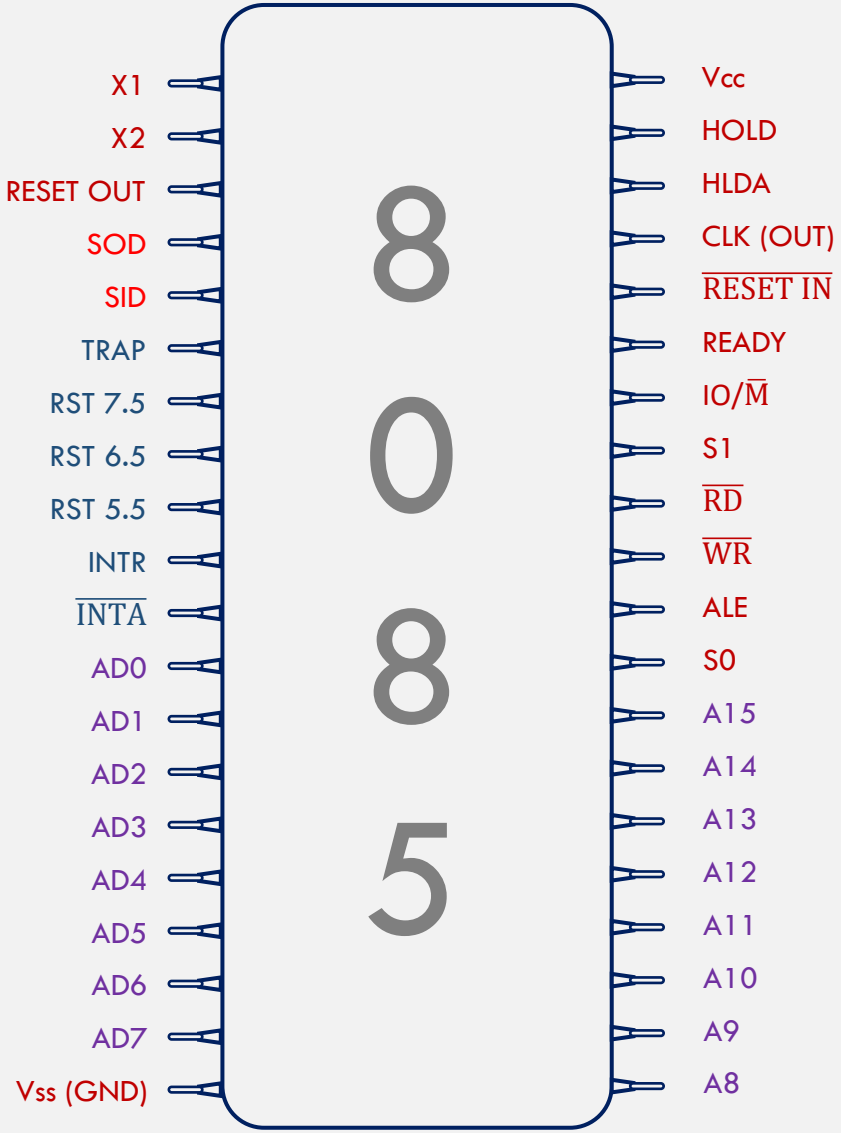
2.6 Interrupt and flags

2.7 Instructions types and data flow inside 8085

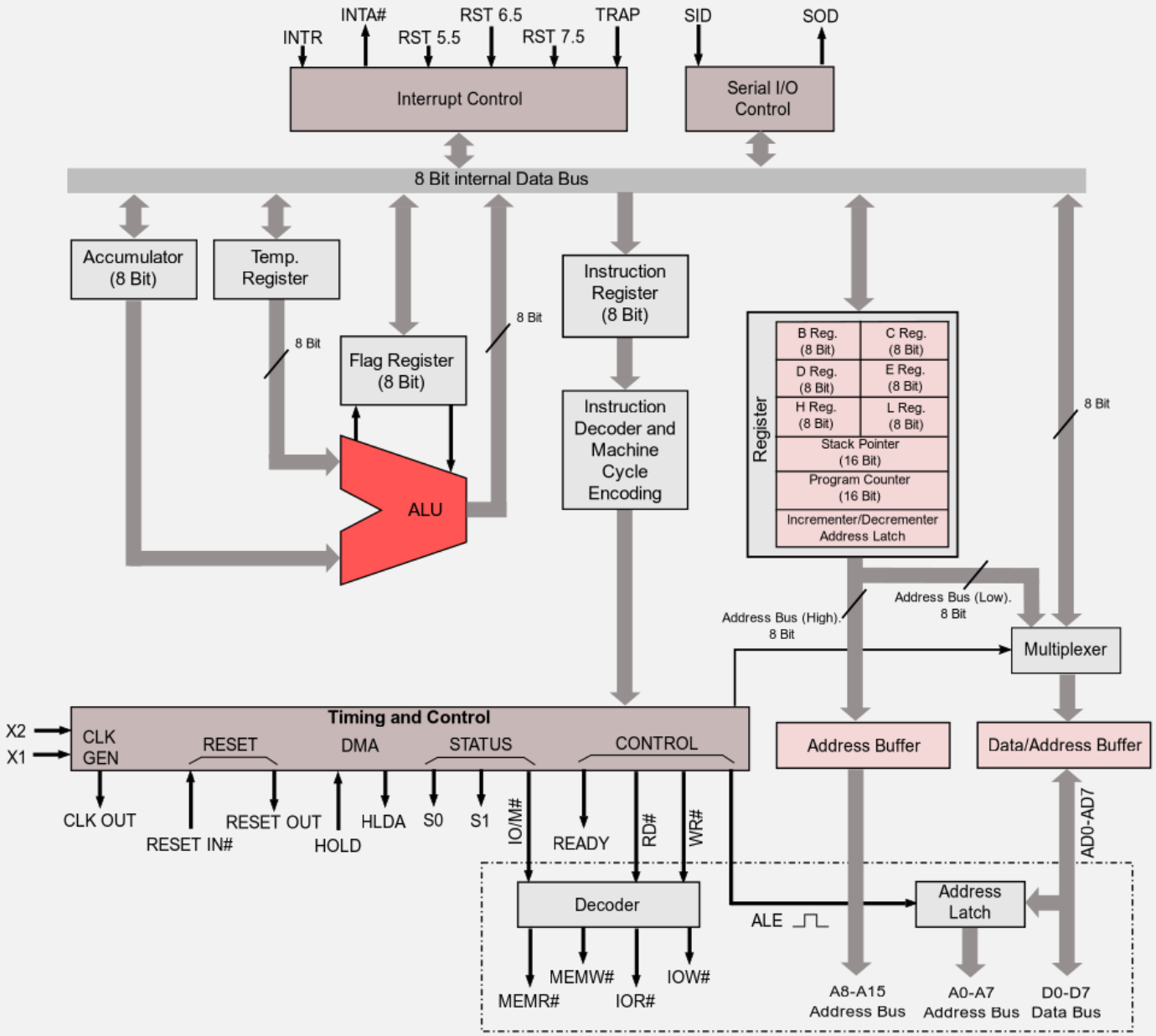
2.8 Timing diagram (two examples of 4,7,10 and 13 t-states)

2.9 Basic Assembly language programming using 8085 instruction sets

Pin diagram of 8085



Functional block diagram of 8085



Intel 8085 instruction types

8- bit	0 or 8 or 16- bit
Opcode	Operand

1. One-byte Instruction Formats

8- bit	0- bit
Opcode	Operand

ADD B, SUB C

2. Two-byte Instruction Formats

8- bit	8- bit
Opcode	Operand

OUT 8-bit addr.

3. Three-byte Instruction Formats

8- bit	16- bit
Opcode	Operand

LDA 16-bit addr.

Intel 8085 instruction types

Opcode Format

Code	Registers
000	B
001	C
010	D
011	E
100	H
101	L
111	A
110	Reserved for Memory related operation

Code	Registers Pairs
00	BC
01	DE
10	HL
11	AF OR SP

Intel 8085 instruction types

"It has 74 operation codes that results in 246 instructions. "

A. Data transfer instructions

Mnemonics	Examples	Operation	
MVI R, 8-bit	MVI B, 7CH	B	7C
	MVI C, 3FH	C	3F
MOV Rd, Rs	MOV B, C	B	
		C	8E
	MOV E, L	E	
		L	4B

Intel 8085 instruction types

A. Data transfer instructions

Mnemonics	Examples	Operation	
LXI Rp, 16-bit	LXI H, 9C2FH	H	9C
		L	2F
	LXI D, 345EH	D	34
		E	5E
OUT 8-bit	OUT 01H	PORT 01	
		A	6F
IN 8-bit	IN 02H	A	
		PORT 02	8E

Intel 8085 instruction types

A. Data transfer instructions

Mnemonics	Examples	Operation			
STA 16-bit	STA 5051H	M	5051		
			A	6F	
LDA 16-bit	LDA 2020H		A		
		M	2020	8E	
LDAX Rp	LDAX B		A		
		M	7C3F	8E	
			B	7C	
			C	3F	

Intel 8085 instruction types

A. Data transfer instructions

Mnemonics	Examples	Operation		
STAX Rp	STAX D	M	3C3E	
		A	2E	
		D	3C	
		E	3E	
MOV R, M	MOV C, M	C		
		M	3D21	8E
		H	3D	
		L	21	

Intel 8085 instruction types

A. Data transfer instructions

Mnemonics	Examples	Operation		
MOV M, R	MOV M, E	M	1D1F	
		E	2E	
		H	1D	
		L	1F	
XCHG	XCHG	D	12	
		E	8E	
		H	54	
		L	22	

Intel 8085 instruction types

B. Arithmetic Instructions

Mnemonics	Examples	Operation
LHLD 16-bit add.	LHLD 5052	

H	D8
L	E6

Memory

5051	27
5052	E6
5053	D8
5054	53

Mnemonics	Examples	Operation
SHLD 16-bit add.	SHLD 2051	

H	A1
L	C2

Memory

2051	C2
2052	A1
2053	
2054	

Intel 8085 instruction types

Data transfer Questions

Write instructions for transferring data from memory location 5051H to B-register.

Write instructions for transferring data from Port 2BH to L-register.

Write instructions for transferring data from memory location 2A05H to 2B05H.

Write instructions for saving data of HL register to memory location 78C5H.

Write instructions for saving data from DE register to memory location 5689H.

Intel 8085 instruction types

B. Arithmetic Instructions

Mnemonics	Examples	Operation
ADD R	ADD C	

A	38
C	15

ALU
3E+15

S	Z		AC		P		CY
0	0		0		0		0

Mnemonics	Examples	Operation
ADI 8-bit	ADI 43H	

A	21

ALU
21+43

S	Z		AC		P		CY
0	0		0		0		0

Intel 8085 instruction types

B. Arithmetic Instructions

Mnemonics	Examples	Operation
ADD M	ADD M	

A	29

H	50
L	53

Memory

5051	27
5052	E6
5053	D8
5054	53

ALU
5129

S	Z		AC		P		CY
0	0		0		0		0

Intel 8085 instruction types

B. Arithmetic Instructions

Mnemonics	Examples	Operation
ADC R	ADC E	

A	34
E	15

ALU
3E+15+1

S	Z		AC		P		CY
0	0		0		0		0

Mnemonics	Examples	Operation
ACI 8-bit	ADI 43H	

A	21

ALU
21+43+1

S	Z		AC		P		CY
0	0		0		0		0

Intel 8085 instruction types

B. Arithmetic Instructions

Mnemonics	Examples	Operation
ADC M	ADC M	

A	29

H	50
L	53

Memory

5051	27
5052	E6
5053	D8
5054	53

ALU
51+D8+0

S	Z		AC		P		CY
0	0		0		0		0

Intel 8085 instruction types

B. Arithmetic Instructions

Mnemonics	Examples	Operation
SUB R	SUB B	
SUI 8-bit	SUI 73H	

A	4B
B	34

ALU
4F-34

S	Z		AC		P		CY
0	0		0		0		0

A	2C

ALU
2B-73

S	Z		AC		P		CY
0	0		0		0		0

Intel 8085 instruction types

B. Arithmetic Instructions

Mnemonics	Examples	Operation
SUB M	SUB M	

A	29

H	20
L	01

Memory

2001	45
2002	96
2003	E1
2004	65

ALU
2E45

S	Z		AC		P		CY
0	0		0		0		0

Intel 8085 instruction types

B. Arithmetic Instructions

Mnemonics	Examples	Operation
INR R	INR D	
INR M	INR M	

D	40

S	Z		AC		P		CY
0	0		0		0		0

H	F0
L	07

Memory

F005	45
F006	96
F007	E2
F008	65

S	Z		AC		P		CY
0	0		0		0		0

Intel 8085 instruction types

B. Arithmetic Instructions

Mnemonics	Examples	Operation
DCR R	DCR B	
DCR M	DCR M	

B	5D

S	Z		AC		P		CY
0	0		0		0		0

H	00
L	06

Memory

0005	4A
0006	06
0007	33
0008	F5

S	Z		AC		P		CY
0	0		0		0		0

Intel 8085 instruction types

B. Arithmetic Instructions

Mnemonics	Examples	Operation
INX Rp	INX B	
DCX Rp	DCX D	

B	50
C	02

D	3D
E	0D

Intel 8085 instruction types

C. Logical and Bit Manipulation Operation

Mnemonics	Examples	Operation
ANA R	ANA B	

A	3D
B	22

ALU
3D&22

S	Z		AC		P		CY
0	0		1		0		0

ANI 8-bit	ANI 74H

A	2F

ALU
2F&74

S	Z		AC		P		CY
0	0		1		1		0

Intel 8085 instruction types

C. Logical and Bit Manipulation Operation

Mnemonics	Examples	Operation
ANA M	ANA M	

A	51

H	50
L	53

Memory

5051	27
5052	E6
5053	D8
5054	53

ALU
51 & D8

S	Z		AC		P		CY
0	0		1		1		0

Intel 8085 instruction types

C. Logical and Bit Manipulation Operation

Mnemonics	Examples	Operation
ORA R	ORA E	

A	47
E	1D

ALU
47 |5|F1 D

S	Z		AC		P		CY
0	0		0		0		0

ORI 8-bit	ORI 6DH

A	32

ALU
32 |7|F6D

S	Z		AC		P		CY
0	0		0		0		0

Intel 8085 instruction types

C. Logical and Bit Manipulation Operation

Mnemonics	Examples	Operation
ORA M	ORA M	

A	37

H	30
L	41

Memory

3041	27
3042	E6
3043	D8
3044	53

ALU
31 β∨27

S	Z		AC		P		CY
0	0		0		0		0

Intel 8085 instruction types

C. Logical and Bit Manipulation Operation

Mnemonics	Examples	Operation
XRA R	XRA E	

A	27
E	56

ALU
71

S	Z		AC		P		CY
0	0		0		0		0

XRI 8-bit	XRI 43H

A	EC

ALU
EF

S	Z		AC		P		CY
0	0		0		0		0

Intel 8085 instruction types

C. Logical and Bit Manipulation Operation

Mnemonics	Examples	Operation
XRA M	XRA M	

A	62

H	50
L	54

Memory

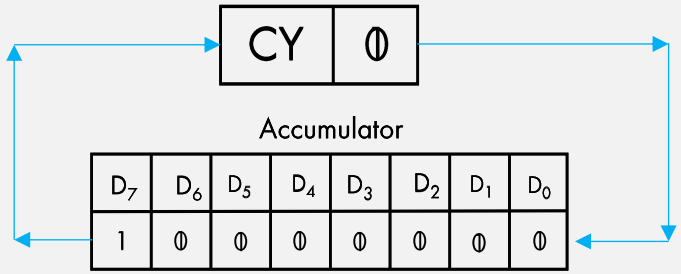
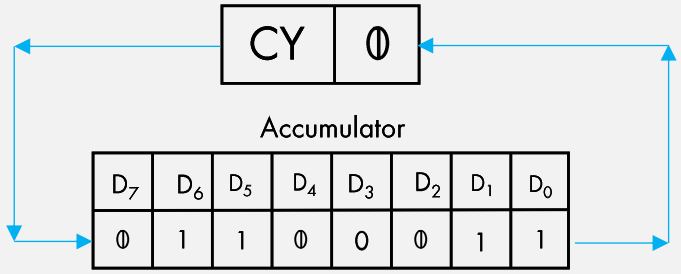
5051	27
5052	E6
5053	D8
5054	53

ALU
5102

S	Z		AC		P		CY
0	0		0		0		0

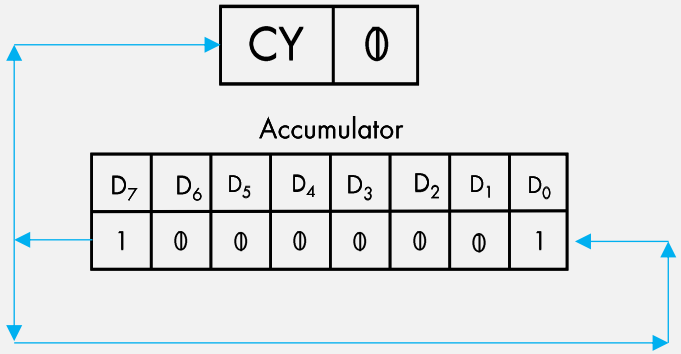
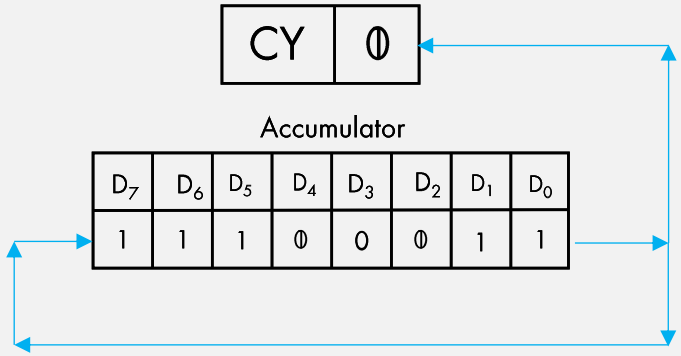
Intel 8085 instruction types

C. Logical and Bit Manipulation Operation

Mnemonics	Examples	Operation
RAL	RAL	
RAR	RAR	

Intel 8085 instruction types

C. Logical and Bit Manipulation Operation

Mnemonics	Examples	Operation
RLC	RLC	
RRC	RRC	

Intel 8085 instruction types

C. Logical and Bit Manipulation Operation

Mnemonics	Examples	Operation
CMA	CMA	
CMC	CMC	

A	3D2

ALU
3D2

S	Z		AC		P		CY
0	0		0		0		0

Intel 8085 instruction types

C. Logical and Bit Manipulation Operation

Mnemonics	Examples	Operation
CMP R	CMP B	

A	71
B	56

S	Z		AC		P		CY
0	0		0		1		0

CMP D

A	71
D	8F

S	Z		AC		P		CY
1	0		0		1		1

CMP C

A	71
C	71

S	Z		AC		P		CY
0	1		1		1		0

Intel 8085 instruction types

Arithmetic and logical instructions Questions

Write instructions for adding data from memory location 5051H and B-register.

How can you find out whether the data of accumulator has 0 or 1 in D_5 location?

Write instructions for multiplying 5 and 4 and store the result in memory location 5050H.

What will be the status of flags if you subtract 78H from 56H?

Write instructions for moving lower nibble to higher nibble.

Intel 8085 instruction types

D. Branch Instructions

Mnemonics	Examples	Operation
JMP 16-bit addr.	JMP 5676H	Change the program sequence to specified 16-bit addr.
JZ 16-bit addr.	JZ FF34H	Change the program sequence to specified 16-bit addr. if the zero flag is set
JNZ 16-bit addr.	JNZ 74FFH	Change the program sequence to specified 16-bit addr. if the zero flag is reset
JC 16-bit addr.	JC EE34H	Change the program sequence to specified 16-bit addr. if the carry flag is set
JNC 16-bit addr.	JNC AAFFH	Change the program sequence to specified 16-bit addr. if the carry flag is reset

Intel 8085 instruction types

D. Branch Instructions

Mnemonics	Examples	Operation
CALL 16- bit addr.	CALL 5676H	Change the program sequence to location of a subroutine
RET	RET	Return to the calling program after completing the subroutine sequence

Intel 8085 instruction types

E. Stack and Machine Control Instructions

Mnemonics	Examples	Operation
HLT	HlT	Stop processing and wait
NOP	NOP	Do not perform any operation
POP Rp	POP H	

SP	3022

H	A1
L	76

Memory

3021	27
3022	76
3023	A1
3024	53

Intel 8085 instruction types

E. Stack and Machine Control Instructions

Mnemonics	Examples	Operation
PUSH Rp	PUSH D	

SP	3021

D	A1
E	76

Memory

3021	76
3022	A1
3023	F1
3024	53

Intel 8085 instruction types

Questions

Make a loop that runs “NOP” instruction for five times.

Multiply 10 and 20.

Multiply 50 and 70.

Count numbers of 1’s in the byte located at 7070H and save the result in 5051H.

Check the number at 8081H is odd or even. Save 1 at 5051H if odd else save 0.

```
MVI A, 00H
MVI B, 14H
MVI C, 0AH

FIRTA_JA:
ADD B
DCR C
JNZ FIRTA_JA
HLT
```

```
MVI C, 05H
FIRTA_JA: NOP
DCR C
JNZ FIRTA_JA
HLT
```

Operation code and operands

An instruction is a command to perform certain task

1. YO 6 NUM KO LYAUNUS HAI
2. MALAI NI TYHI LYAUNUS
3. TYO KETCHUP LYAU TA
4. ABA KHAM HAI

Addressing modes

A. Immediate Addressing Mode

Immediate value is given in an instruction.

Examples: MVI A, 55H , ADI 4FH

A	55

B. Register Addressing Mode

Data to be operated is present in a register

Examples: MOV C,A , ADD D

B	
C	8E

C. Direct Addressing Mode

Memory location of data is directly specified on instruction

Examples: LDA 505FH , STA 5454H

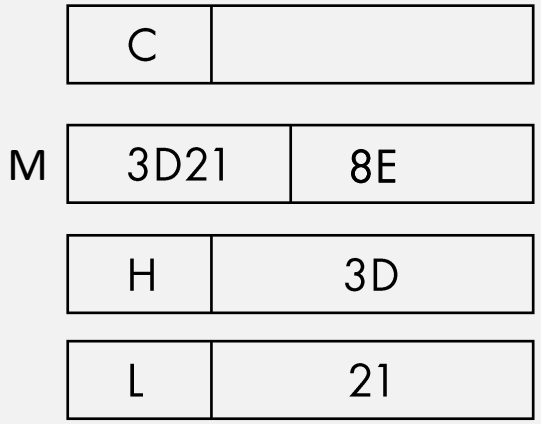
	A	
M	2020	8E

Addressing modes

D. Register Indirect Addressing Mode

Memory location of data is found in register specified on instruction

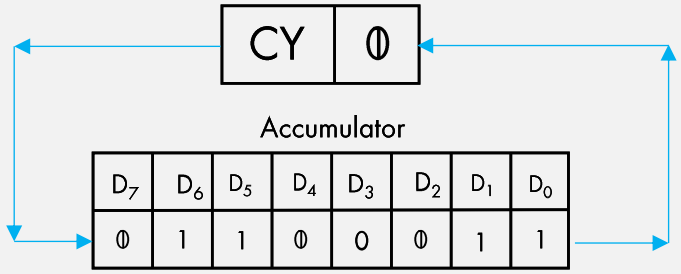
Examples: MOV B,M , ADD M



E. Implied Addressing Mode

Microprocessor already knows data to be operated

Examples: RAR , RLC



Timing Diagram

It is the graphical representation of step by step working mechanism of microprocessor with respect to time.

It represents read, write and status signals.

A. Instruction Cycle

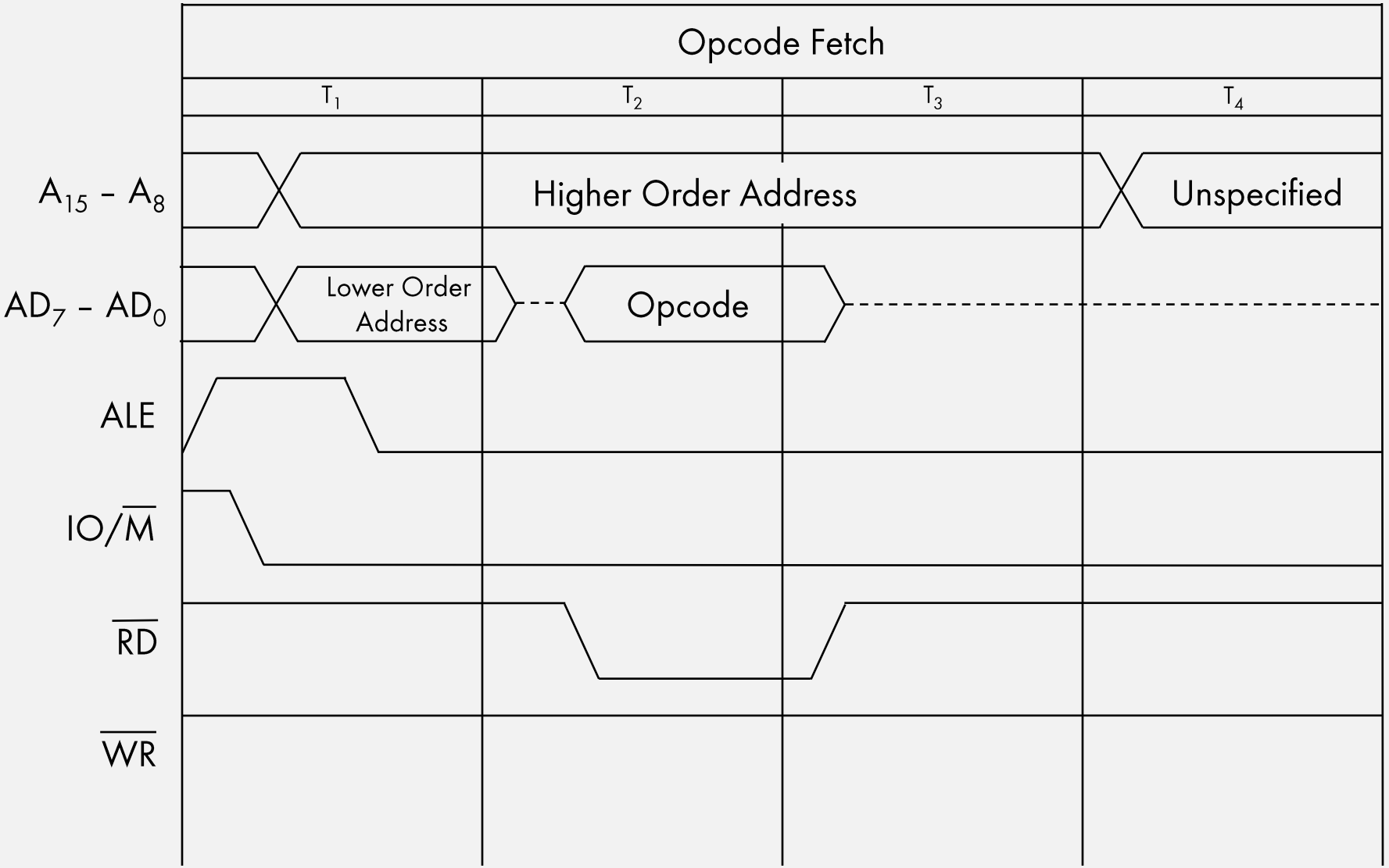
Total number of steps required by CPU to complete an instruction. Fetching, decoding and execution of an instruction.

B. Machine Cycle

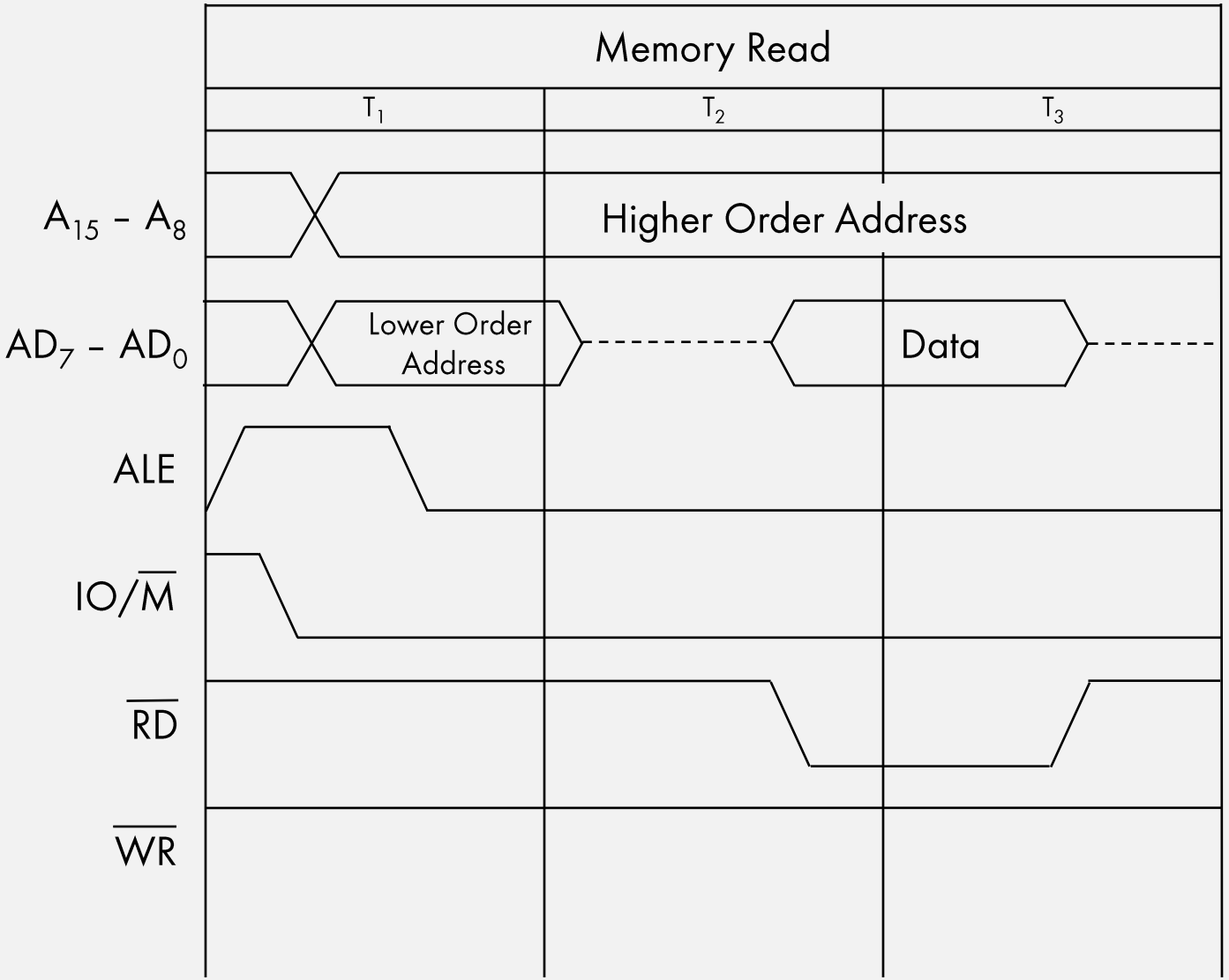
Total number of steps required by CPU to complete the operation of accessing memory or I/O.

Fetch, memory read, memory write, I/O read, I/O write.

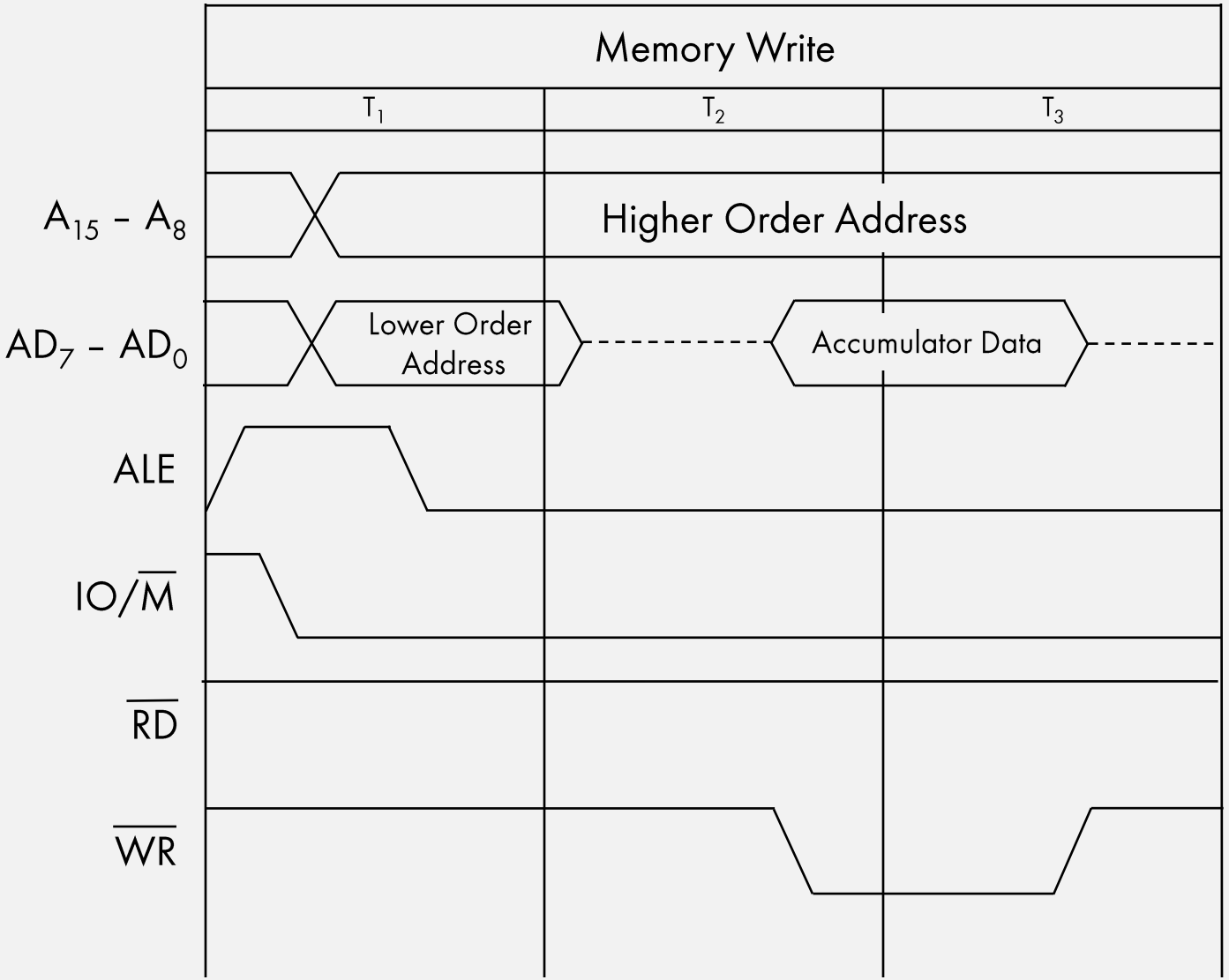
Timing Diagram



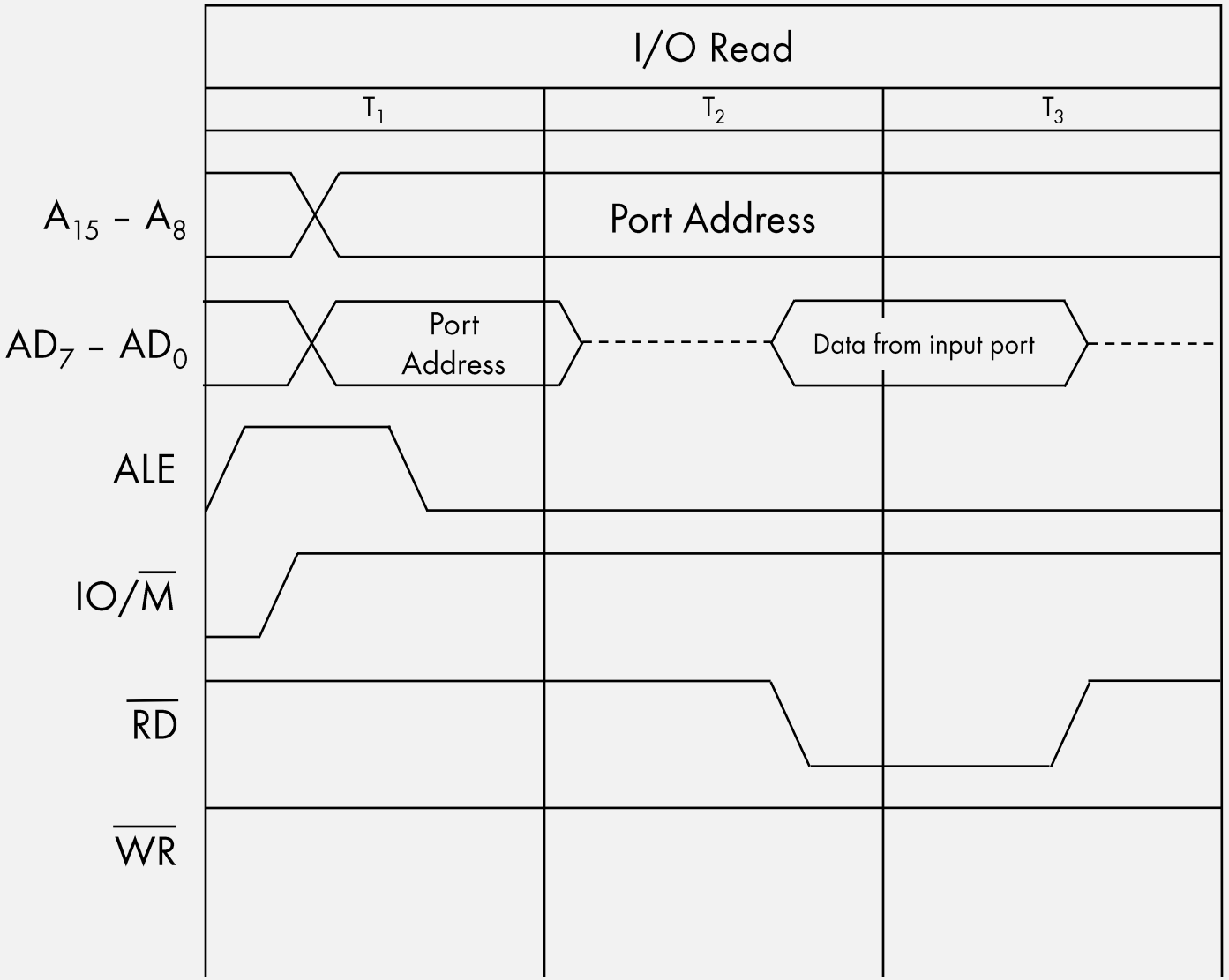
Timing Diagram



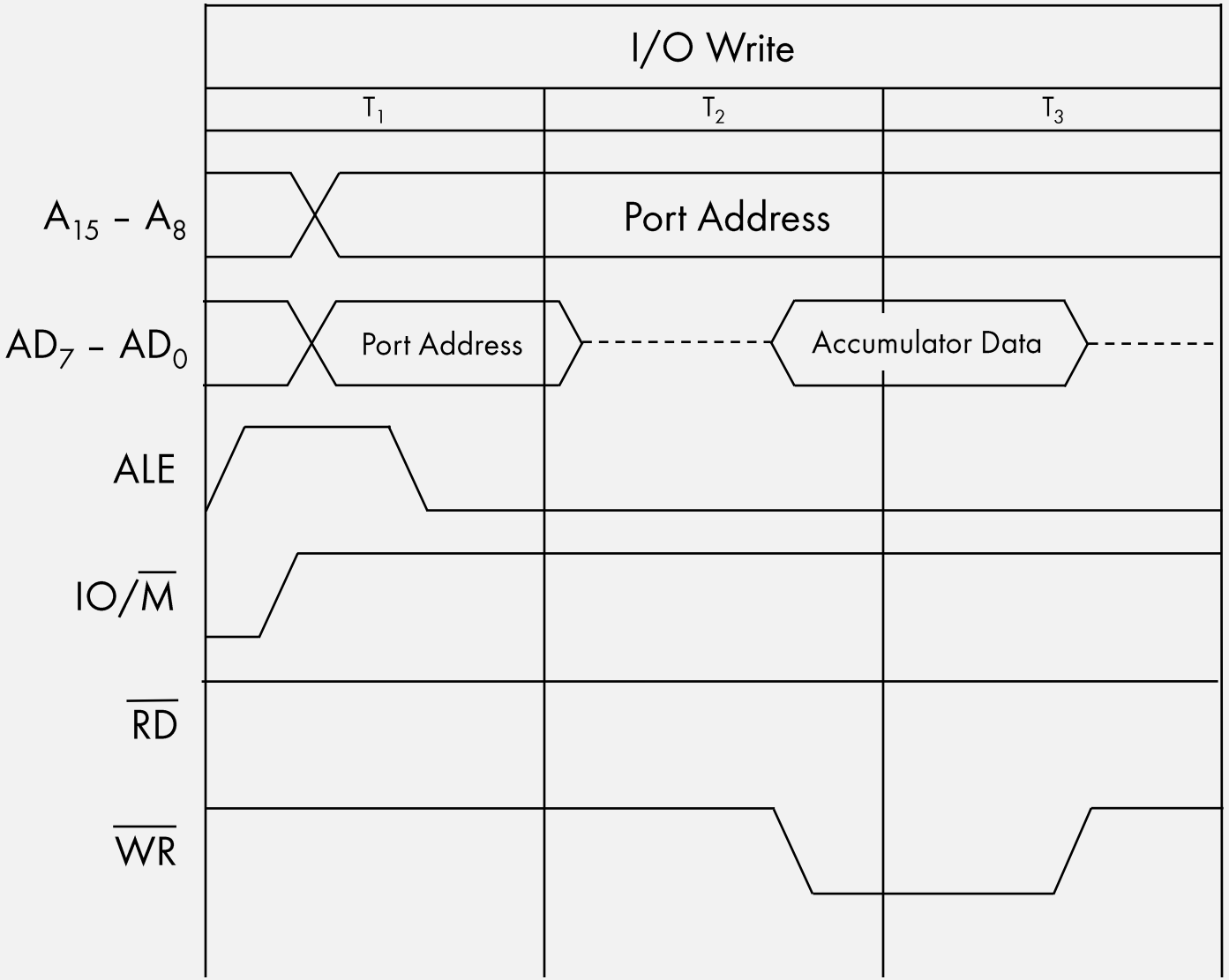
Timing Diagram



Timing Diagram



Timing Diagram



Timing Diagram

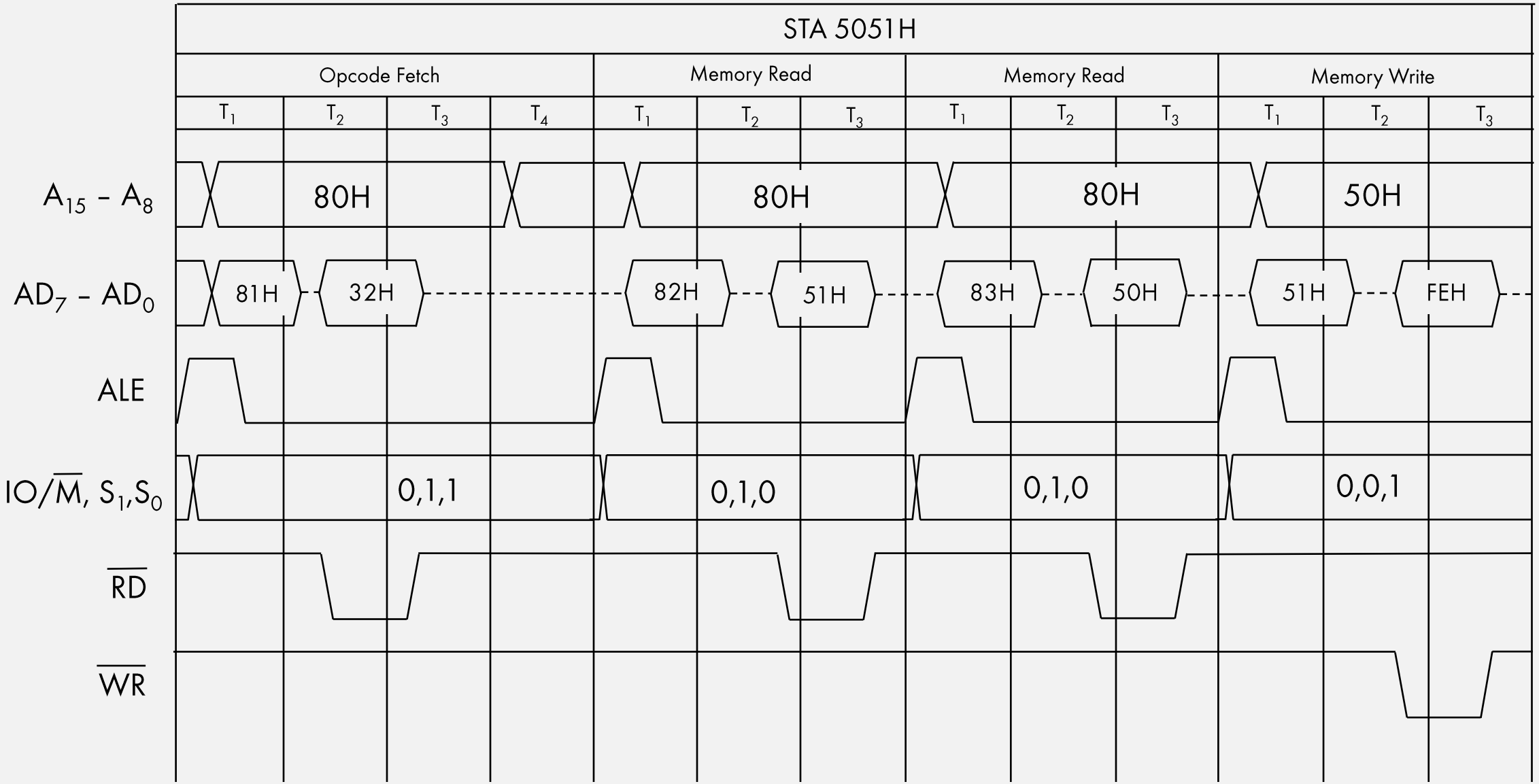
Draw a timing diagram of an instruction STA 5051. Assume starting memory address as 8081H and content of accumulator as FEH.

A	11111110	FE

Opcode Fetch	ME. Read	ME. Read	ME. Write
4-T States	3-T States	3-T States	3-T States

Memory Address	Memory Content	H
.	XXXXXXXX	XX
.	XXXXXXXX	XX
5051	XXXXXXXX	XX
5052	XXXXXXXX	XX
.	XXXXXXXX	XX
.	XXXXXXXX	XX
.	XXXXXXXX	XX
8081	00110010	32
8082	01010001	51
8083	01010000	50
8084	XXXXXXXX	XX
.	XXXXXXXX	XX
.	XXXXXXXX	XX
.	XXXXXXXX	XX

Timing Diagram



Timing Diagram

Draw a timing diagram of an instruction IN 01H. Assume starting memory address as 3031H and content of input device as 54H.

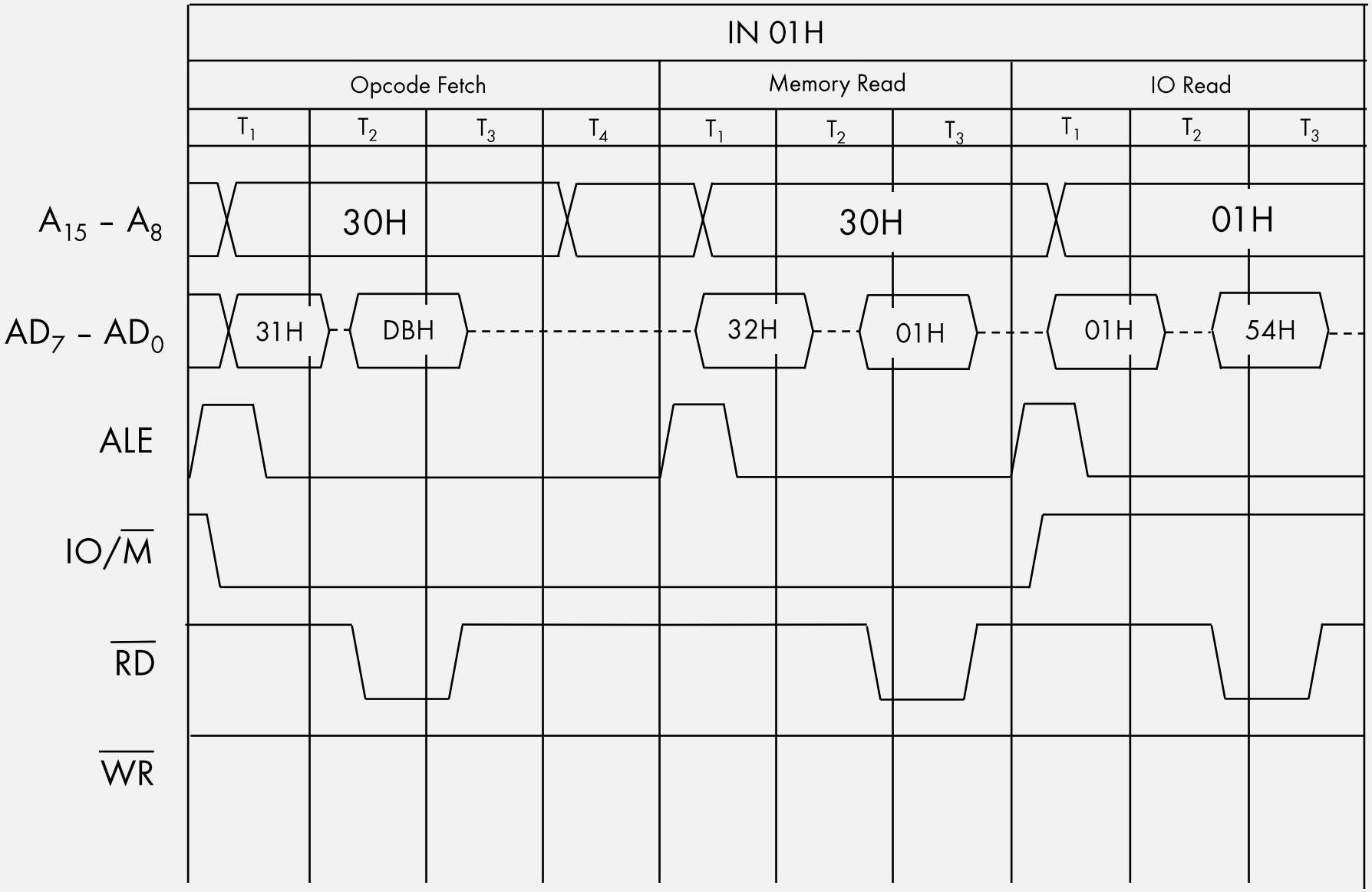
PORT 01	01010100

A		54

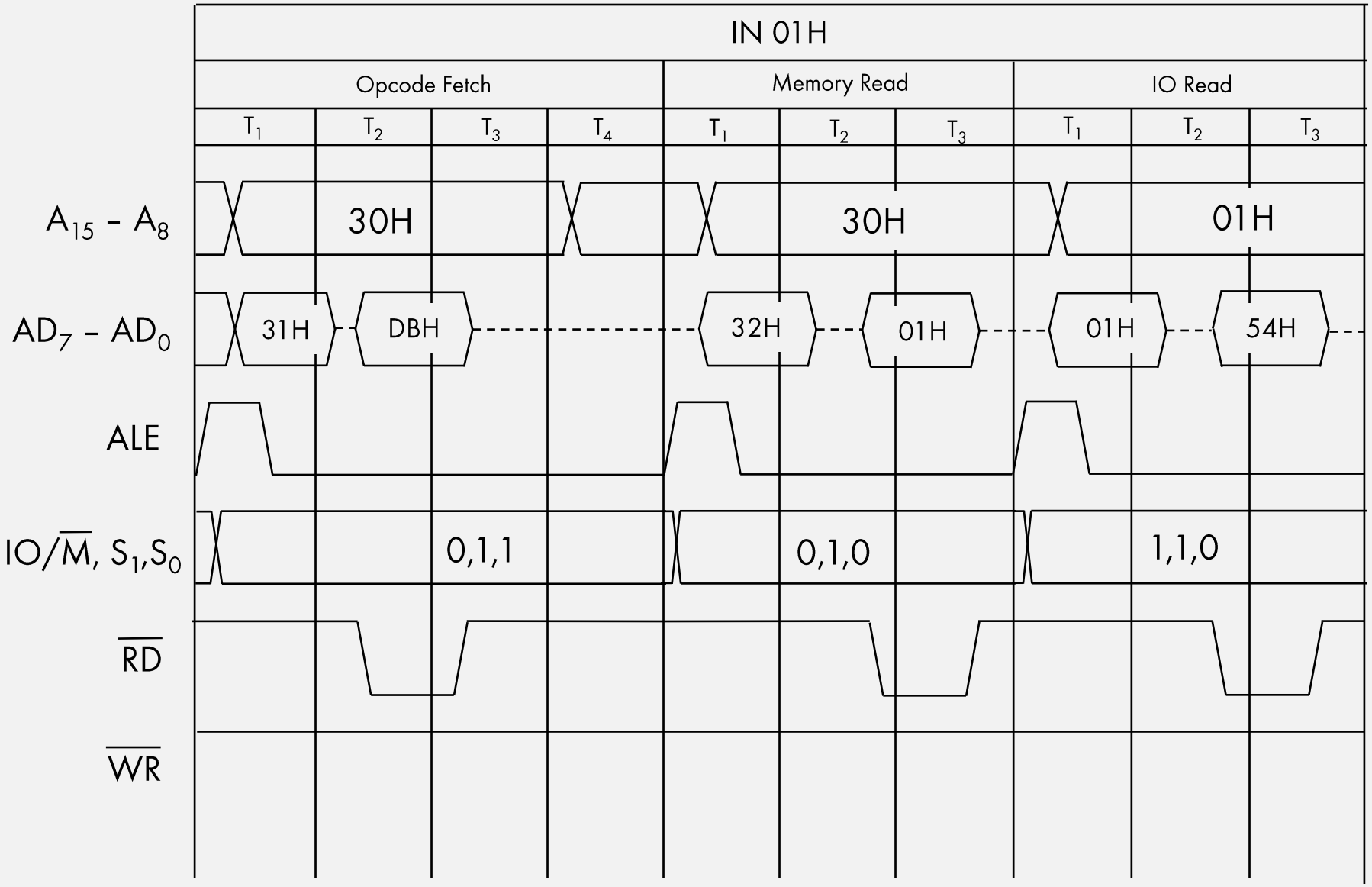
Opcode Fetch	ME. Read	IO Read
4-T States	3-T States	3-T States

Memory Address	Memory Content	H
.	XXXXXXXX	XX
.	XXXXXXXX	XX
.	XXXXXXXX	XX
.	XXXXXXXX	XX
.	XXXXXXXX	XX
.	XXXXXXXX	XX
.	XXXXXXXX	XX
3031	11101011	DB
3032	00000001	01
3033	XXXXXXXX	XX
3034	XXXXXXXX	XX
.	XXXXXXXX	XX
.	XXXXXXXX	XX
.	XXXXXXXX	XX

Timing Diagram



Timing Diagram



Status signal in 8085

S_1	S_0	Operation
0	0	Halt
0	1	Write
1	0	Read
1	1	Fetch